

Section-A (25 Marks)

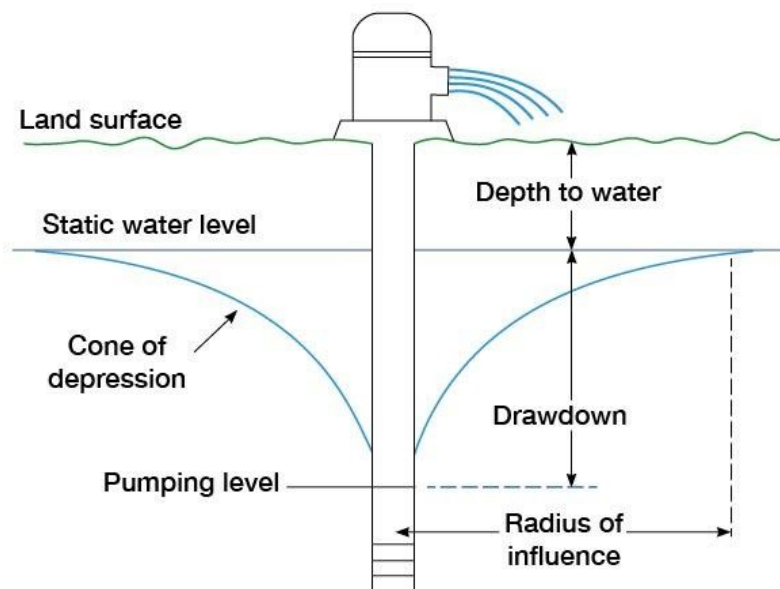
1. What do you understand by inverted cone of depression and circle of influence in well? Name the different method of tube well boring and indicate the conditions where each one is suitable. Sketch and explain geological condition which give rise to an artesian well. [10]

Answer:

Inverted cone of depression and circle of influence in well are the terms used to describe the phenomenon of lowering the ground water table when the water is pumped.

Inverted cone of depression:

- It is a cone-shaped lowering of groundwater table that develops around a pumping well.
- When a well starts pumping, water levels near the well drop, creating a zone of drawdown that is deepest at the well and gradually less with distance.
- If the aquifer is homogeneous and no strong natural gradient exists, the cone is approximately symmetrical and circular in plan, with radial flow toward the well.



Circle of influence:

- It is the imaginary circular boundary surrounding a pumping well within which the groundwater level is affected by pumping.
- In plan view, the cone of depression appears as a series of concentric drawdown contours. The outermost contour where drawdown becomes negligible indicates the circle of influence in the well.
- The size of this circle depends on pumping rate, aquifer transmissivity, duration of pumping, and boundary conditions.

The method of tube well boring is selected based on the type of soil formation, depth of well, required diameter and groundwater conditions. The different method of tube well boring are listed below:

i. **Auger boring:**

- It is suitable for soft clay, silt, sandy clay and loam soil above the water table.
- Used for shallow wells up to about 15-20 m.

ii. **Wash boring:**

- It is suitable for loose sandy, silty and clayey soils below the water table.
- It is commonly used for small diameter tube wells.

iii. **Percussion boring:**

- It is suitable for hard clay, gravel, boulders, weathered rock and hard formations.
- This method can drill hard wells but is slower.

iv. Rotary drilling:

- It is suitable for deep tube wells.
- Direct rotary drilling is best for soft to medium soil formations.
- Reverse rotary drilling is preferred for large-diameter wells in unconsolidated alluvial formations.

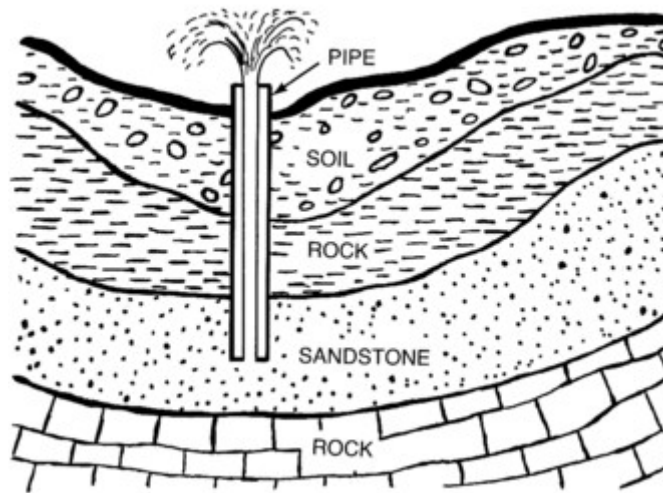
v. Driven well method:

- It is suitable for very shallow wells in loose sand or sandy gravel where groundwater is available at shallow depth.

An artesian well is a type of well that taps into a confined aquifer, also known as a pressure or artesian aquifer. In this geological setup, groundwater is confined under pressure greater than atmospheric pressure. If the pressure is sufficient, the water may flow out at the ground surface without pumping; this is called a flowing artesian well.

Geological conditions which give rise to artesian well:

An artesian well forms under the following geological conditions:



i. Confined Aquifer

- A permeable water-bearing layer such as sandstone, limestone, or fractured rock is confined between two impermeable layers such as clay or shale.
- This confined layer stores groundwater under pressure.

ii. Recharge Area at Higher Elevation

- The confined aquifer must be exposed at the surface in a higher elevation area where rainfall infiltrates into it.
- Water enters the aquifer through this recharge zone.

iii. Impermeable Confining Beds

- The aquifer is overlain and underlain by impermeable strata that prevent vertical escape of water.
- These layers trap the groundwater, creating pressure.

iv. Hydrostatic Pressure

- The recharge area is at a higher elevation than the well site, so that groundwater in the confined aquifer is under pressure.

- When a well penetrates the aquifer, water rises to the piezometric surface.
- If the piezometric surface lies above the ground level, the well flows naturally.

2. a. Explain the procedure for site selection of permanent intake when the water levelling fluctuating to large extent during wet and dry seasons. [10]

Answer:

Intake is the structure that help in safely withdrawing of water from the source and discharge that water to intake conduit through which it flows to the supply line. The main function of the intakes work is to collect water from the water source and then discharge water so collected.

The site for locating intake should be selected considering the following points.

- i. The location of the intake site should be as far as possible near to the treatment plant so that the cost of conveying would be less.
- ii. Water quality available in the intake site should be good which will reduce the load on treatment plant.
- iii. The intake site should be located to ensure the water supply even in the worst condition.
- iv. The intake site should be located such that there is sufficient scope for future expansion and additions.
- v. The intake site should be easily accessible even during the floods.
- vi. The intake site should be located geologically stable and free from possibilities of erosion, silting and scouring.

The water level in the natural sources fluctuates seasonally. During wet seasons, the water level reduces due to increased consumption and extraction of water while during wet seasons, the water level increases due to rainfall, surface runoff and infiltration.

Hence, to overcome the fluctuation of water level to large extent during wet and dry season, the procedure for the site selection of permanent intake are as follows:

- i. **Hydrological Investigation**
 - The long-term river flow records are studied.
 - Highest flood level (HFL) and lowest water level (LWL) are determined.
 - The seasonal discharge variation and drought flow are assessed.
- ii. **Selection of Stable River Reach**
 - A straight, stable section of the river are chosen.
 - The sharp bends, confluences, and braided channels where erosion or sediment deposition is excessive.
 - It must be ensured that the river course is unlikely to shift.
- iii. **Adequate Water Depth**
 - A location is selected where sufficient depth is available even during the dry season.
 - If necessary, a weir or barrage is provided to maintain the required water level.
- iv. **Flood Safety**
 - The intake must be located above the maximum flood level or provide adequate flood protection.
 - Intake structures must be designed to withstand flood currents and floating debris.
- v. **Sediment Consideration**
 - Locations with heavy silt deposition or excessive bed load must be avoided.
 - Sites with relatively clear flow is preferred.
 - Provision of sediment exclusion and desilting arrangements is ensured.
- vi. **Water Quality Assessment**
 - Upstream location must be selected away from:
 - Sewage outfalls
 - Industrial discharges
 - Agricultural runoff
 - Bathing and washing
 - The source providing good-quality raw water is selected.

vii. **Geological and Foundation Conditions**

- A site with firm, stable foundations such as rock or dense soil must be chosen.
- Landslide-prone, highly erodible, or unstable banks must be avoided.

viii. **Accessibility and Economy**

- Easy access for construction, operation, and maintenance is ensured.
- The locations close to the treatment plant is preferred to reduce pipeline length and pumping costs.

ix. **Future Expansion**

- Sufficient space for additional intake units or increased water demand in the future.

3. **What are the different types of distribution service reservoir used in water supply project in Nepal? How its storage capacity is determined? Discuss in brief. [10]**

Answer:

Distribution service reservoir is the type of storage structures that are designed to meet balancing reserve, breakdown reserve, and firefighting reserve. Service reservoir are used in the distribution system to provide storage:

- To balance the variations of water demand in a day.
- To provided storage for firefighting.
- To manage emergencies like breakdown, repairs etc.
- To stabilize the pressure in the distribution systems.

According to the ground conditions, service reservoirs are classified as:

i. Surface reservoirs:

- These are rectangular or circular in shape and constructed at the ground level or below the ground level.
- They are also known as ground reservoirs or non-elevated reservoirs.
- Treated water from this reservoir is lifted to elevated reservoir to supply consumers or supplied directly if surface reservoirs located at higher elevations.



ii. Elevated reservoirs:

- They may be circular, rectangular, circular, egg-shaped, spherical or elliptical in shapes constructed at an elevation from the ground.
- The assorted accessories provided for elevated reservoir are manhole, inlet, outlet, overflow, ladder etc.



iii. Standpipes:

- A vertical cylindrical tank resting just above the ground up to 15 to 30 m height and diameter 10 to 15 m which are made of steel or RCC standpipes.
- The purpose of such tank is to increase pressure in the distribution system by creating extra storage in the tank above the elevation required to give the necessary pressure for the distribution of water.



Storage capacity of these service reservoirs are determined by determination of:

- i. Balancing reserve:
- ii. Breakdown storage or emergency storage:
- iii. Fire storage:

The total of above three categories determines the capacity of the service reservoir.

i. Balancing reserve:

- This storage is provided to meet the fluctuating demand of water with a constant rate of pumping of water into the reservoir.
- It can be determined by:

a. Hydrograph method:

- In this method, hourly demands of water for a typical maximum day are plotted against the respective hours of the day and a hydrograph is obtained.
- For a uniform 24 hours of pumping, the pumping rate will be equal to the mean hourly demand of water and the same is plotted by line AB.
- The storage required for a distribution reservoir is then obtained by determining the area between the curve QRST and the line AB.
- The area covered into volume units gives the storage required for balancing reservoir.
- Analytical method is more convenient for determining the storage reservoir capacity for a distribution system.

b. Mass curve method:

- In this method, a mass curve is drawn which is a plot of cumulative demand of water against time.
- Values of cumulative demand of water at successive hours are obtained and are plotted against the corresponding hours to obtain the mass curve of demand. A mass curve continuously rises.
- Draw tangents through the lowest point E and highest point F. Find the two ordinates between demand and supply curves and calculate $S=A+B$.

ii. Breakdown storage:

- The breakdown storage requirements may be due to the failure of pumps, power supply, repair time etc.
- It is difficult to assess the magnitude of the storage to meet this requirement hence 25% of the total capacity of the reservoir or about 1.5 to 2 times the average hourly supply may be considered as breakdown storage.

iii. Fire storage:

- The water stored in service reservoir to provide water for firefighting purpose is fire reserve.
- Water required for fire storage can be determined by:

$$FR = [F - P] \times T$$

where,

FR= Fire reserve

F= Fire demand

P= Reserve for pumping capacity

T= Duration of fire

4. It is required to supply water to a population of 20000 at a per capita demand of 150 litres per day. The disinfectant used for chlorination is much bleaching powder is required annually at the water works if 0.3 pm of chlorine dose is required. [10]

Answer:

Population, $P=20,000$

Per capita water demand, $q=150$ L/person/day

Chlorine dose $=0.3$ ppm $=0.3$ mg/L

Assume bleaching powder contains 30% available chlorine.

Total water demand, $Q = P \times q$

$$Q = 20,000 \times 150$$

$$Q = 3,000,000 \text{ L/day}$$

Since chlorine dose $= 0.3$ mg/L,

$$\text{Daily chlorine required} = 3,000,000 \times 0.3$$

$$= 900,000 \text{ mg / day}$$

$$= \frac{900,000}{10^6} = 0.9 \text{ kg / day}$$

$$= 106900,000 = 0.9 \text{ kg / day}$$

Annual chlorine requirement= $0.9 \times 365 = 328.5 \text{ kg / year}$

Bleaching powder contains 30% available chlorine.

$$\text{Bleaching powder required} = \frac{328.5}{0.30}$$

$$= 1095 \text{ kg / year}$$

The annual quantity of bleaching powder required is:

$$\approx 1.10 \text{ tonnes / year}$$

5. **Outline the importance of community participation for sustainability of the water supply project. Point out the challenges. [10]**

Answer:

The active involvement of local people in the planning, design, implementation, operation, maintenance, monitoring, and decision-making of a water supply project is community participation. It is a key factor for the long-term sustainability of water supply systems especially in the developing country like Nepal.

Importance of community participation:

- i. **Creates a sense of ownership:**
 - People feel responsible for protecting and maintaining the system.
- ii. **Improves operation and maintenance:**
 - Communities can promptly identify and repair minor faults, reducing system downtime.
- iii. **Ensures sustainability:**
 - Local participation helps keep the project functional long after construction is completed.
- iv. **Mobilizes local resources:**
 - Communities contribute labor, local materials, land, or financial support, reducing project costs.
- v. **Better planning and decision-making:**
 - Local knowledge helps in selecting suitable water sources, intake locations, and distribution systems.
- vi. **Enhances transparency and accountability:**
 - Community involvement reduces misuse of funds and promotes proper management.
- vii. **Improves hygiene and sanitation awareness:**
 - Communities become more aware of safe water handling, sanitation, and public health practices.
- viii. **Strengthens local institutions:**
 - Formation of Water Users Committees improves local management capacity.
- ix. **Reduces conflicts:**
 - Joint decision-making minimizes disputes over water allocation and system operation.

Challenges of community participation:

- i. Lack of awareness and education about water management.

- ii. Limited technical knowledge for operation and maintenance.
- iii. Insufficient financial resources for repairs and maintenance.
- iv. Weak leadership and poor institutional capacity.
- v. Political interference in project management.
- vi. Social inequality, leading to exclusion of women, poor households, or marginalized groups.
- vii. Conflicts among users over water distribution and responsibilities.
- viii. Low willingness to contribute labor or maintenance fees.
- ix. Migration of active community members, reducing local participation.
- x. Poor coordination between the community and government agencies.

6. a. what is a self-cleansing velocity and non-scouring velocity in a sewer? Why sewers are not designed to a full flow? [4]

B. Design a sewer to serve a population of 36000; the daily per capita water supply allowance being 135 liters, of which 80% find its way into the sewer. The slope available for the sewer to be laid is 1 in 625 and the sewer should be designed to carry 4 times the dry weather flow when running full. What would be the velocity of flow in the sewer when running full? Assume $n=0.012$ in Manning's formula.

Answer:

a. Self- cleansing velocity:

- It is the minimum velocity at which the solid particles remains in suspension without settling at sewer's bottom.
- It is the velocity at which even scour of the deposited particles of the given size will take place.
- It must be maintained at least once a day as it is not possible to maintain the velocity throughout the day.

Formula for Self-Cleansing Velocity:

According to Shield's Theory:
$$V_s = \sqrt{\frac{8 K (\rho_s - \rho) g d}{f \rho}}$$

Where:

V_s = Self-cleansing velocity (m/s)

K = Dimensionless constant

ρ_s = Density of solid particles (kg/m³)

ρ = Density of water (kg/m³)

g = Acceleration due to gravity (9.81 m/s²)

d = Diameter of particle (m)

f = Friction factor

Non-scouring velocity in the sewer:

- It is the maximum velocity at which interior sewer surface is not damaged.
- It is the velocity at which no scour or abrasion takes place.
- Non-Scouring Velocity prevents erosion of channel bed and banks and ensures stability of sewers.

Formula for Non-Scouring Velocity

According to Kennedy's Theory:

$$V_0 = 0.55 m D^{0.64}$$

Where:

V_0 = Non-scouring (critical) velocity (m/s)

m = Critical velocity ratio

D = Depth of flow (m)

For average silt conditions:

$$m = 1.0$$

Therefore,

$$V_0 = 0.55 D^{0.64}$$

Sewers are not designed for full flow conditions because:

i. Variation of sewage flow:

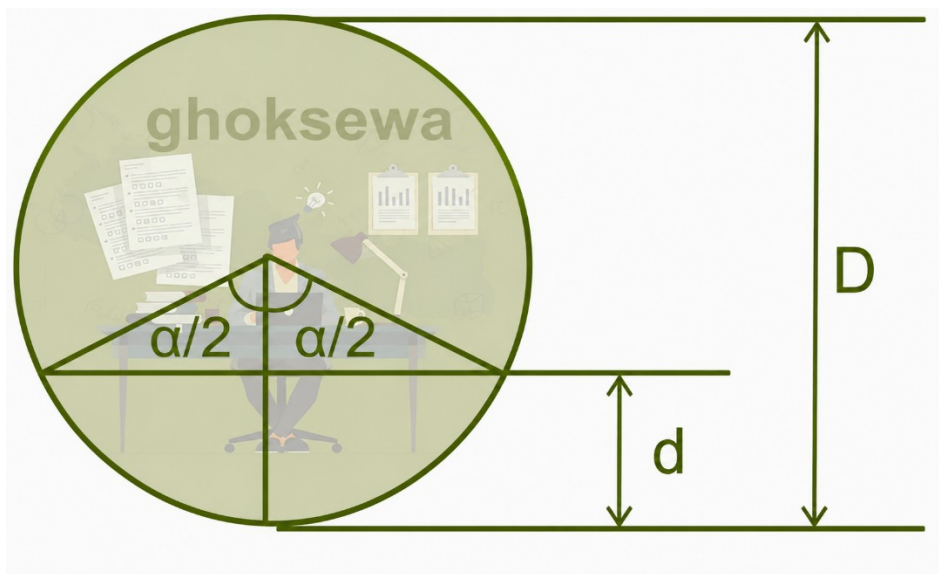
Sewage flow varies throughout the day. Hence, to cater the peak sewage flow during storms, infiltration and future population growth, sewer is designed for the partial flow conditions.

ii. Ventilation:

The space left in partially flowing condition in sewer allows air circulation which prevents accumulation of harmful and foul gases such as Hydrogen Sulphide gas, methane gas etc.

iii. Prevention of overflow:

If sewer runs in full flow condition, slight increase in discharge causes overflowing which results in unhygienic condition in the area.



b.
Given,

$$\text{Population, } P = 36,000$$

$$\text{Per capita water supply} = 135 \text{ L / person / day}$$

$$\text{Sewage contribution} = 80\% \text{ of water supply}$$

$$\text{Sewer slope, } S = \frac{1}{625} = 0.0016$$

Sewer designed to carry Dry Weather Flow (DWF) when running full

$$\text{Manning's coefficient, } n = 0.012$$

$$\text{DWF} = 36000 \times 135 \times 0.80$$

$$= 3,888,000 \text{ L / day}$$

$$\text{Convert to } m^3/s:$$

$$Q_{DWF} = \frac{3888}{86400} = 0.045 \text{ m}^3/\text{s}$$

The sewer is designed for four times the DWF.

$$Q = 4 \times 0.045 = 0.18 \text{ m}^3/\text{s}$$

For a circular sewer running full,

$$Q = \frac{1}{n} A R^{2/3} S^{1/2}$$

For a full circular pipe,

$$A = \frac{\pi D^2}{4}$$

$$R = \frac{D}{4}$$

Substituting,

$$Q = \frac{1}{0.012} \left(\frac{\pi D^2}{4} \right) \left(\frac{D}{4} \right)^{2/3} (0.0016)^{1/2}$$

Since

$$\sqrt{0.0016} = 0.04$$

This simplifies to

$$Q = 2.638 D^{8/3}$$

Putting $Q = 0.18$,

$$0.18 = 2.638 D^{8/3}$$

$$D^{8/3} = 0.06823$$

$$D = (0.06823)^{3/8} = 0.366 \text{ m}$$

Hence,

$$D \approx 0.37 \text{ m}$$

Adopt the nearest standard diameter:

$$D = 375 \text{ mm}$$

Using,

$$V = \frac{1}{n} R^{2/3} S^{1/2}$$

For the calculated diameter, $D = 0.366 \text{ m}$,

$$R = \frac{0.366}{4} = 0.0915 \text{ m}$$

$$V = \frac{1}{0.012} (0.0915)^{2/3} (0.0016)^{1/2}$$

$$V = 83.33 \times 0.203 \times 0.04$$

$$V \approx 0.68 \text{ m/s}$$

7. **Sewage Treatment Plant मध्ये oxidation pond को फाइदाको बारेमा उल्लेख गर्नुहोस् | [10]**

Answer:

Oxidation pond is an open pond, flowing through earthen basins, designed and constructed for the treatment of sewage and industrial waters.

Detention period in oxidation pond is longer ranging from few to several days.

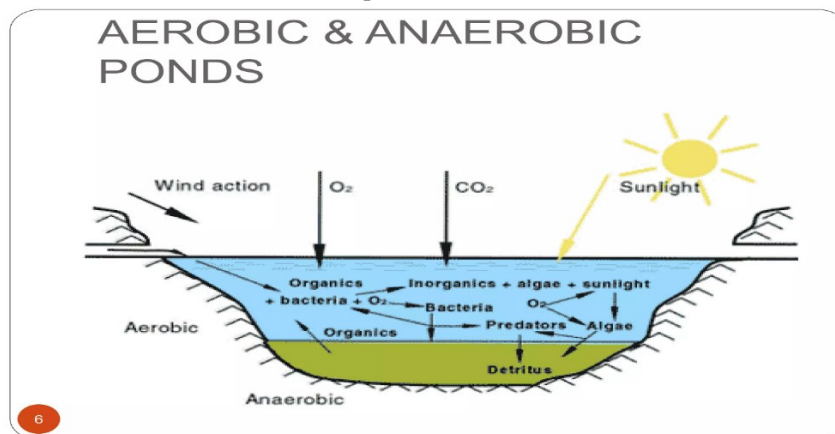
Oxidation pond is used to treat municipal sewage as well as biodegradable industrial sewage.

Stabilization of sewage is done by aerobic, anaerobic as well as facultative bacteria in oxidation pond. There are three zones created in oxidation pond. They are:

- i. Aerobic zone
- ii. Anaerobic zone
- iii. facultative zone

Functions of oxidation pond:

- i. To settle the organic matter at the bottom of the pond.
- ii. To reduce coliform present in the sewage.
- iii. To reduce the load on the treatment plant.



For the design of oxidation pond, the data we require are listed below:

- i. Pond shape:
There should not be any corners or extended portion.
Shape may be round, square or any regular shape with the length not exceeding 3 times the width.
- ii. Embankment:
Top width=1.5-3 m'
Freeboard= 0.5 m (minimum)
Inner slope= 1:1 to 1.5:1
Outer slope= 2:1 to 2.5:1
- iii. Detention period= 10-45 days
- iv. Depth of oxidation pond= 1-1.5 m
- v. Surface area < 0.5 hectares. If > 0.5 hectares, multiple ponds can be constructed.
- vi. Loading criteria:
For cold region= 55 kg/ha/ day
For tropic region= 300 kg/ha/day

8. **Describe the different stages in sludge digestion process briefly. What are the different factors affecting sludge digestion and how they are controlled? Do you think sludge disposal by lagooning in case of Nepal is appropriate and why?**

Answer:

The biological decomposition of organic matter in sewage sludge under anaerobic conditions is known as sludge digestion. It reduces the volume of sludge, stabilizes it, destroys pathogens, and produces biogas such as methane.

The sludge digestion process occurs in four main stages:

i. Hydrolysis:

- Complex organic matter such as carbohydrates, proteins, and fats is broken down into simpler soluble compounds by hydrolytic bacteria.
- This is the first and often rate-limiting stage.

Examples:

Carbohydrates → Simple sugars

Proteins → Amino acids

Fats → Fatty acids + Glycerol

ii. Acid formation/ Acidogenesis:

- The soluble compounds are converted by acid-forming bacteria into:
 - Volatile fatty acids (VFAs)
 - Alcohols
 - Carbon dioxide (CO₂)
 - Hydrogen (H₂)
 - Ammonia (NH₃)
- The pH tends to decrease due to acid production.

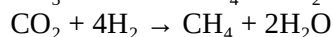
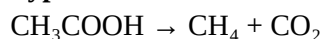
iii. Acetate formation/Acetogenesis:

- Volatile fatty acids and alcohols are further converted into:
 - Acetic acid (CH₃COOH)
 - Carbon dioxide (CO₂)
 - Hydrogen (H₂)
- These products serve as substrates for methane-forming bacteria.

iv. Methane formation/ Methanogenesis:

- Methanogenic bacteria convert acetic acid, hydrogen, and carbon dioxide into:
 - Methane (CH₄)
 - Carbon dioxide (CO₂)

Typical reactions:



This stage stabilizes the sludge and produces biogas, which can be used as a renewable energy source.

The efficiency of anaerobic sludge digestion depends on several physical, chemical, and biological factors. Proper control of these factors ensures stable digestion, maximum gas production, and well-stabilized sludge. The factors affecting the sludge digestion and their control measures are given below:

i. Temperature:

- Microbial activity increases at high temperature.
- Control measures:
Constant temperature is maintained. Mesophilic bacteria are maintained at 30–38°C, usually 35°C and thermophilic bacteria at 50–57°C.

ii. pH:

- Low pH inhibits methanogenic bacteria and high pH also reduces bacterial activity.
- Control measures:
pH is maintained between 6.8 and 7.5 by proper buffering and balanced loading.

iii. Organic loading:

- Overloading of sludge causes acid accumulation and process failure while underloading reduces efficiency.
 - Control measures:
Feed sludge is fed at a uniform and optimum loading rate.
- iv. Retention time:
- Insufficient retention results in incomplete digestion.
 - Control measures:
Adequate hydraulic retention time (HRT) typically 15–30 days for mesophilic digestion is provided.
- v. Mixing:
- Poor mixing causes scum formation and uneven digestion.
 - Control measures:
The mechanical mixing, gas recirculation, or sludge recirculation is provided to ensure uniform conditions.

Yes, sludge disposal by lagooning is appropriate in many parts of Nepal, particularly for small and medium-sized towns and rural areas, but it is not suitable for densely populated urban areas due to land constraints and environmental concerns. It can be used in Nepal because:

- i. The method is simple and economical as it requires low construction, operation, and maintenance costs.
- ii. It is suitable for municipalities with limited financial resources.
- iii. The method does not require complex machinery or high electricity consumption.
- iv. Nepal's moderate to warm temperatures in many regions promote natural drying and biological stabilization.
- v. Lagooning can be managed with limited technical skills.
- vi. Sunlight, evaporation, and microbial action reduce moisture and stabilize the sludge.
- vii. After proper drying and stabilization, sludge obtained from lagooning can be used as a soil conditioner or fertilizer.



9. **Define eco-sanitation. What can you suggest socio-economically poor village people of disposal septic tank effluent, whereas they are facing with their hand-to-mouth problems? [10]**

Answer:

Eco-sanitation, or ecosan, is a sanitation approach that treats human excreta and wastewater as resources which is safely collected, treated, and reused rather than disposed of. It protects the public health, helps to conserve water, and recover nutrients for agriculture.

For the economically poor village households facing “hand-to-mouth” conditions, expensive treatment systems are usually not feasible. The following low-cost, easily manageable and practical options are recommended:

- i. Soak pit:

- Discharge septic tank effluent into a soak pit.
- Allows natural infiltration and further treatment through the soil.
- Low construction and maintenance cost.

ii. Leach field:

- Distribute effluent through perforated pipes laid in gravel-filled trenches.
- Provides additional treatment by soil microorganisms.
- Suitable where adequate land is available.

iii. Oxidation pond:

- Suitable where sufficient land is available.
- Uses natural biological processes and sunlight to treat wastewater.

iv. Reuse of treated sludge:

- Safe reuse of treated sludge/compost on non-food crops only after proper treatment and drying, if local acceptance and hygiene controls are ensured.

10. **a) Explain the role of CO₂ on greenhouse effect. [5]**

b) Why IEE is necessary for water supply project. [5]

Answer:

a) Carbon dioxide (CO₂) is one of the greenhouse gases in the earth's atmosphere. It makes up only about 0.04% of the atmosphere and plays a major role in regulating the earth's temperature. While methane is 40 times more potent than carbon dioxide, CO₂ remains the single largest driver of global warming, accounting for roughly 75 % to 80% of the total long-term warming effect caused by human greenhouse gas emissions.

Role of CO₂ on greenhouse effect:

- Sun emits short-wave radiation, which passes through the atmosphere and reaches the Earth's surface.
- The Earth's surface absorbs this solar energy, warms up, and emits long-wave infrared (heat) radiation.
- CO₂ molecules absorb part of this outgoing infrared radiation instead of allowing it to escape directly into space.
- CO₂ re-emits the absorbed heat in all directions, including back toward the Earth's surface.
- This traps heat in the lower atmosphere, warming the Earth.

b.

- According to Environment Protection Act (EPA) 2076 and Environment Protection Rules (EPR) 2077, IEE is defined as "An analytical study or evaluation conducted to determine whether a proposed project or activity will cause adverse environmental impacts during its implementation, and to identify measures to avoid, reduce, or mitigate such impacts."
- IEE is conducted to determine whether a proposed medium scale project or activity will cause adverse environmental impacts during its implementation, and to identify measures to avoid, reduce, or mitigate such impacts.
- If IEE Report submitted is approved by concerned authority, the project is implemented.
- IEE is done for:
 - a. Construction and operation of water treatment plants with a capacity exceeding **100 Liters Per Second**

Initial Environment Examination (IEE) is required for water supply projects because:

i. **Identifies Environmental Impacts**

- Assesses impacts on water sources, forests, land, air, noise, and biodiversity during construction and operation.

ii. **Protects Water Sources and public health:**

- Ensures sustainable withdrawal of water without depleting rivers, springs, or groundwater.
- Evaluates risks of water contamination and recommends measures to ensure safe drinking water.
- iii. **Suggests Mitigation Measures**
 - Recommends actions to minimize adverse impacts, such as erosion control, proper waste disposal, and source protection.
- iv. **Promotes Sustainable Resource Use**
 - Encourages efficient use and conservation of water resources for present and future generations.
- v. **Ensures Legal Compliance**
 - Helps comply with the **Environment Protection Act, 2076** and **Environment Protection Rules, 2077** of Nepal (or other applicable environmental regulations).
- vi. **Protects Biodiversity and Ecosystems**
 - Prevents damage to forests, wildlife habitats, wetlands, and aquatic ecosystems.
- vii. **Facilitates Public Participation**
 - Provides opportunities for consultation with affected communities and stakeholders, increasing project acceptance.